Square roots over n
Rudin, 3.7

Theorem 1. *If $\sum a_n$ converges and $a_n \geq 0$ for every n, then $\sum \frac{\sqrt{a_n}}{n}$ converges.*

Proof. We consider the series indexed by $n = 1, 2, \ldots$.

The p-series with $p = 2$ converges, so

$$\sum_n \frac{1}{n^2}$$

converges. Since $\sum a_n$ converges by assumption, it follows that

$$\sum_n \frac{a_n + \frac{1}{n^2}}{2}$$

also converges.

For each n, we have $\sqrt{a_n} \geq 0$ and $n \geq 0$, so

$$0 \leq \frac{\sqrt{a_n}}{n}.$$

Moreover, since $a_n \geq 0$, we have $(\sqrt{a_n})^2 = a_n$. Hence

$$0 \leq \left(\sqrt{a_n} - \frac{1}{n}\right)^2 = (\sqrt{a_n})^2 + \left(\frac{1}{n}\right)^2 - 2\sqrt{a_n} \cdot \frac{1}{n} = a_n + \frac{1}{n^2} - 2\frac{\sqrt{a_n}}{n}.$$

Therefore

$$2\frac{\sqrt{a_n}}{n} \leq a_n + \frac{1}{n^2},$$

and thus

$$\frac{\sqrt{a_n}}{n} \leq \frac{a_n + \frac{1}{n^2}}{2}.$$

Thus $\frac{\sqrt{a_n}}{n}$ is nonnegative and is bounded above by the general term of a convergent series. By the comparison test for nonnegative series, $\sum \frac{\sqrt{a_n}}{n}$ converges. □